

Day 14 – Networking Basics & Troubleshooting

Quick Concepts

OSI vs TCP/IP: OSI has 7 layers; TCP/IP has 4 layers. OSI is conceptual, TCP/IP is practical.

IP → Internet Layer | TCP/UDP → Transport | HTTP/HTTPS & DNS → Application

Example: `curl https://example.com` = Application → TCP → IP

Hands-on Commands & Observations

1. Identity

```
hostname -I
```

Observation: Shows local IP (e.g., 192.168.x.x)

2. Reachability

```
ping google.com
```

Observation: Check latency and packet loss

3. Path

```
tracert google.com
```

Observation: Shows hops between you and destination

4. Ports

```
ss -tulpn
```

Observation: Lists listening services (e.g., SSH on port 22)

5. DNS

```
nslookup google.com
```

Observation: Resolves domain to IP

6. HTTP Check

```
curl -I https://google.com
```

Observation: Returns HTTP status (200 OK = working)

7. Connections

```
netstat -an | head
```

Observation: Shows active and listening connections

Mini Task: Port Check

```
nc -zv localhost 22
```

Observation: If success → port is reachable; else check service/firewall

Reflection

Fastest check: ping

DNS issue: Check Application layer + resolv.conf

HTTP 500: Application issue → check logs